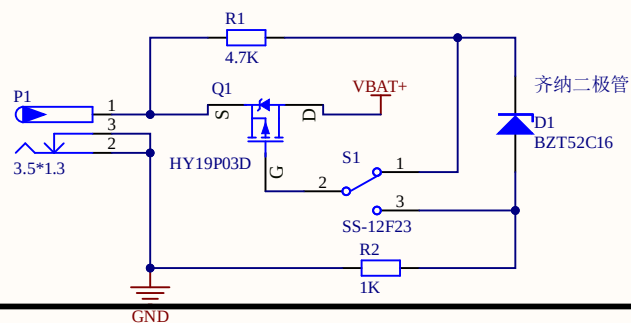
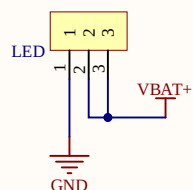


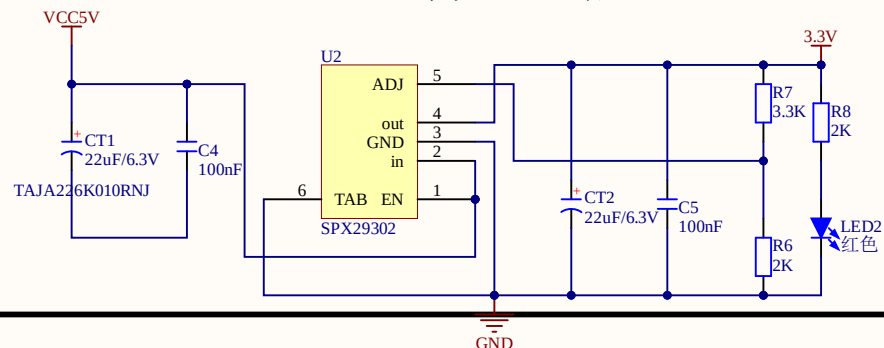
电源输入&保护电路



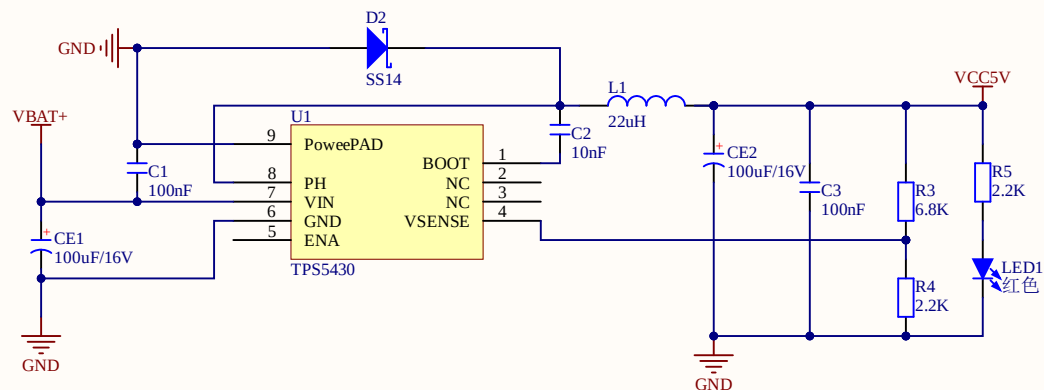
输入电压检测



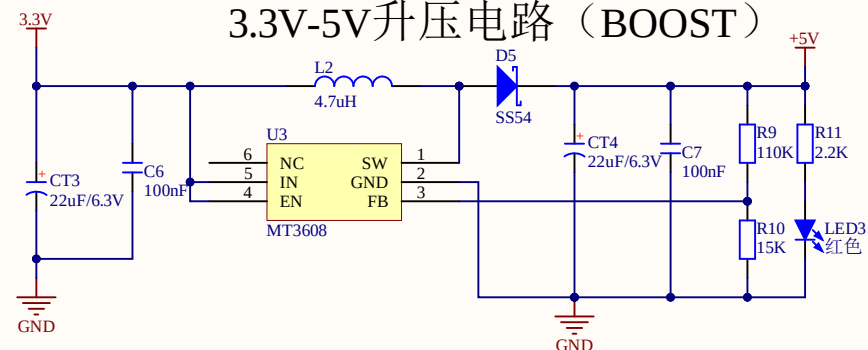
5V-3.3V降压电路 (LDO)



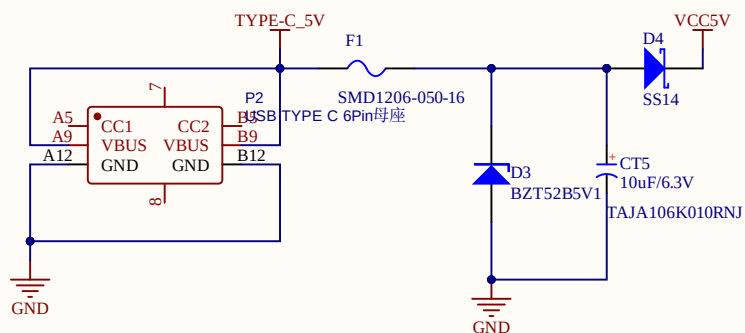
VIN-5V降压电路 (BUCK)



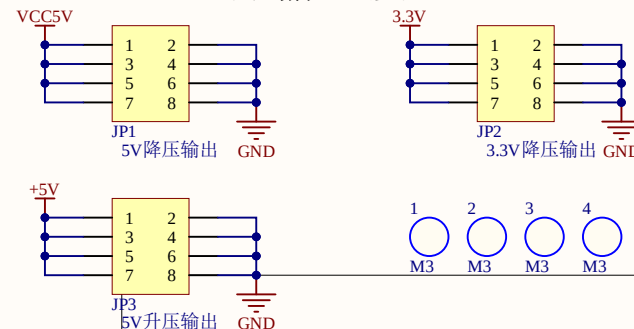
3.3V-5V升压电路 (BOOST)



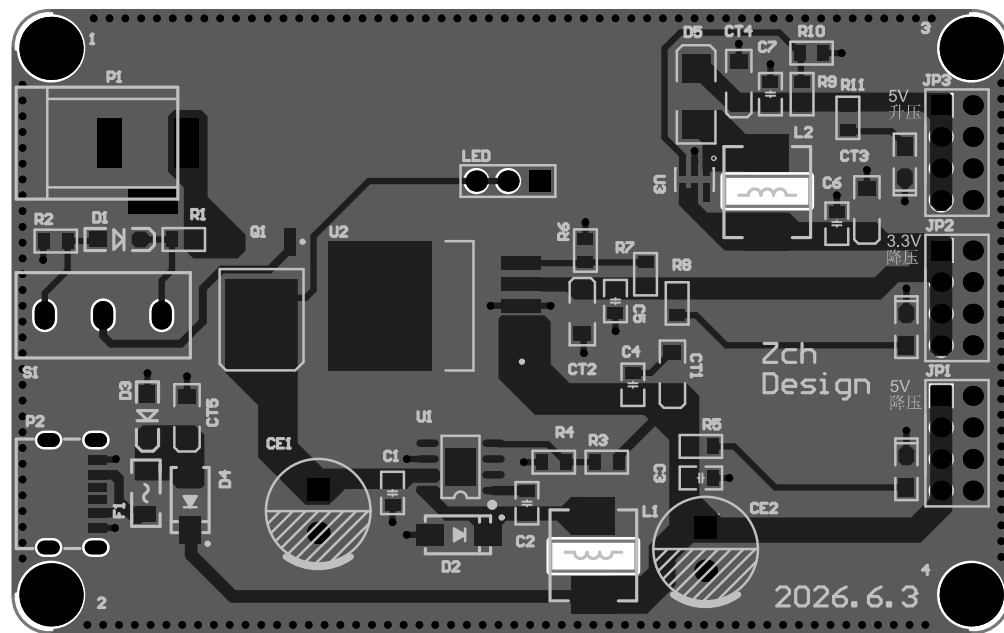
TYPE-C-5V输入



电源输出接口



Size A4	Number	Revision
Date:	7/17/2026	Sheet of
File:	E:\APP\w_dainyuan_SchDoc	Drawn By:



Comment	Description	Designator	Footprint	LibRef	Quantity
M3	铜柱孔	1, 2, 3, 4	M3 125X200	M3	4
100nF	贴片电容	C1, C3, C4, C5, C6, C7	C 0805_L	C	6
10nF	贴片电容	C2	C 0805_L	C	1
100uF/16V	直插电解电容	CE1, CE2	CAP 3.5*8*11.5 - BK	CE	2
22uF/6.3V	贴片钽电容	CT1, CT2, CT3, CT4	CD A(3216)	CT	4
10uF/6.3V	贴片钽电容	CT5	CD A(3216)	CT	1
BZT52C16	Zener Diode (齐纳)	D1	SOD-123	稳压二极管	1
SS14	肖特基二极管	D2, D4	SMA	SS14	2
BZT52B5V1	Zener Diode	D3	SOD-123	稳压二极管	1
SS54	肖特基二极管	D5	SMB	SS54	1
SMD1206-050-16	自恢复保险丝	F1	FU 1206	FU	1
Header 4X2	4*2P接插件	JP1, JP2, JP3	HDR2.54-LI-2x4P	Header 4X2	3
22uH	小功率贴片电感	L1	L-SMD-0630	L	1
4.7uH	小功率贴片电感	L2	L-SMD-0630	L	1
HDR-1X3	3P接插件	LED	HDR2.54-M-LI-3P	Header 3	1
红色	贴片LED	LED1, LED2, LED3	LED 0805R	LED-SMD	3
3.5*1.3	低压电源接口	P1	DC-005	P-DC	1
USB TYPE C 6Pin母座		P2	USB-C-SMD_10P-P1.00-L6.8-W8.9	USB TYPE C 6Pin母座	1
HY19P03D	P沟场效应管	Q1	TO252 - 3	IRF540	1
4.7K	贴片电阻	R1	R 0805_L	R	1
1K	贴片电阻	R2	R 0805_L	R	1
6.8K	贴片电阻	R3	R 0805_L	R	1
2.2K	贴片电阻	R4, R5, R11	R 0805_L	R	3
2K	贴片电阻	R6, R8	R 0805_L	R	2
3.3K	贴片电阻	R7	R 0805_L	R	1
110K	贴片电阻	R9	R 0805_L	R	1
15K	贴片电阻	R10	R 0805_L	R	1
SS-12F23	1路波动开关	S1	SS-12D10	SSW-1P	1
TPS5430	DC可调降压	U1	SOIC-8_EP	TPS5430	1
SPX29302	LDO线性稳压芯片	U2	TO263-5	SPX29302	1
MT3608	DC升压芯片	U3	SOT23-6	MT3608	1